

Peter Min

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. in Mechanical Engineering

May 2029

- GPA: 4.0/4.0
- Relevant Coursework: Calculus I–III; Linear Algebra; Physics: Mechanics and Waves; Physics: Heat, Electricity, and Magnetism; Introduction to Computer Programming; Three-Dimensional Modeling for Design; Thermodynamics; Solid Mechanics; Statistics and Data Science for Engineers

TECHNICAL SKILLS

CAD: SolidWorks, Fusion 360, Onshape

Programming: Python, MATLAB, C++, Git

EXPERIENCE

Undergraduate Researcher

December 2025 – Present

High Performance Robotics Lab, UC Berkeley

Berkeley, CA

- Led manufacturing and component design for an autonomous fixed-wing tiltrotor VTOL aircraft capable of lifting over $4\times$ its own mass.
- Developing flight control software in ROS on a PX4/Pixracer flight controller to achieve stable autonomous flight of the custom-built aircraft.
- Managed the bill of materials and component selection, keeping the full assembly within a strict weight budget.
- Built and maintained the master CAD assembly in Fusion 360, integrating individual components into a complete, buildable model.
- Designed custom component mounts for secure structural connections, enabling modular assembly and rapid replacement of damaged parts.
- Sized primary load-bearing structures (spars, mounts) through hand calculations for bending stress and buckling, ensuring adequate strength at minimum weight.

Propulsion and Simulations Subteam Member

August 2025 – May 2026

Space Enterprise Berkeley

Berkeley, CA

- Supported component assembly and execution of cold-flow testing for a student-built liquid-propulsion rocket targeting the Kármán line (100 km).
- Designed a vertical lift system in Fusion 360 to raise a horizontally resting rocket into launch position, improving test and launch-readiness efficiency.

Intern Researcher

June 2024 – August 2024

Caltech (California Institute of Technology)

Pasadena, CA

- Derived and formally proved a formula for the optimal pebbling number of maximal lobster trees — an original result in combinatorial graph theory.
- Analyzed the structural properties of lobster trees to determine optimal pebbling numbers and their placements.
- Presented findings to Caltech faculty and peer researchers.

Calculus Tutor

May 2023 – June 2024

Self-Employed

Pasadena, CA

- Tutored a student in preparation for the AP Calculus BC exam.
- Developed personalized study plans, practice tests, and targeted review problems to strengthen understanding.

PROJECTS

Rocket-Powered RC Aircraft

Aug. 2026 – Present

- Leading controls, electronics, and sensor integration for a rocket-powered RC aircraft targeting 564 mph (907 km/h) to break the world speed record for a non-jet remote-controlled plane.
- Contributing to component design for a fully 3D-printed airframe.
- Running propulsion calculations for the rocket engine.